

AI-103 Study Guide: Module 1.1 Enhanced

Plan and Prepare to Develop AI Solutions on Azure

This is an ENHANCED version with critical gaps filled in from TXT lecture

1. Exam Domain Mapping

AI-103 Exam Domain	Relevance	What to Remember
Plan and manage an Azure AI solution	High	Foundry project/resource structure, tools, endpoints, SDKs, authentication, responsible AI
Implement generative AI and agentic solutions	Medium	Generative AI, agents, tools, model deployments, Foundry Agent Service
Implement text analysis solutions	Low-Medium	Azure Language, Translator
Implement computer vision solutions	Low-Medium	Vision and multimodal capabilities
Implement information extraction solutions	Low-Medium	Document Intelligence, Content Understanding

Exam focus: Choosing the right Azure AI capability, Microsoft Foundry structure, Foundry Tools, developer tools, authentication approach, and responsible AI considerations.

2. What This Module Teaches

This module teaches how to plan an AI solution on Azure before implementation.

Key idea:

Azure AI solutions are not just model calls. They can combine models, AI services, prompt engineering, tools, agents, custom code, knowledge sources, endpoints, SDKs, monitoring, governance, and responsible AI controls.

Microsoft Foundry is introduced as the **recommended platform for most Azure AI development** because it provides:

- * Project organisation
- * Resource management
- * Model access

- * Agent development
- * Tools integration
- * Endpoints and SDKs
- * Portal experience

3. Key Concepts

3.1 AI Capability Areas

Capability	What It Means	Best-Fit Use	Example
Generative AI and agents	LLM-based apps that generate responses; agents add instructions and tools	Chat apps, task automation, AI assistants	Customer support chatbot
Natural language processing	Text analysis using language models or statistical/task-specific NLP	Entity extraction, sentiment, summarisation	Extract invoice data, analyze reviews
Computer speech	Speech recognition and speech synthesis	Voice apps, transcription, spoken responses	Real-time transcription, voice assistant
Computer vision	Image/video understanding and generation	Image analysis, visual QA, video analysis	Document scanning, facial recognition
Information extraction	Extract key facts from documents, forms, media, and content	Receipts, invoices, forms, document processing	Extract line items from receipts

Exam tip: Know which capability to use for specific scenarios.

3.2 Microsoft Foundry Architecture

Microsoft Foundry is the main Azure platform for building AI apps and agents.

Core Components:

Component	Purpose	What to Know
Foundry portal	Visual interface for projects, models, agents, tools, endpoints, and access	Web-based at https://ai.azure.com
Foundry SDK	Programmatic access for automation, apps, scripts, and CI/CD	Python: <code>azure-ai-projects</code> package
Foundry projects	Organise solution-specific assets	One default project per resource
Foundry resources	Azure resource layer that supports one or more projects	Shared infrastructure layer
Models	Deployed model assets from Foundry Models catalog	Over 1,900 models available
Agents	LLM + instructions + tools	Foundry Agent Service
Tools	Built-in, custom, third-party, MCP, and Foundry Tools	Available through project
Knowledge	Data sources used to ground agents and responses	Via Foundry IQ or vector stores

3.3 Foundry Resource vs Foundry Project

Item	Meaning	Scope	Exam Tip
Foundry resource	Azure-level resource providing shared services, tools, compute, storage, access, and resource endpoint	Shared across projects	Think shared infrastructure
Foundry project	Project-level container for models, agents, connections, knowledge, and project endpoint	Single project workspace	Think solution workspace

Item	Meaning	Scope	Exam Tip
Default project	One project under a resource can be designated as default	Used by tooling/SDKs	Simplifies SDK workflows

WARN CRITICAL DISTINCTION: A Foundry resource and Foundry project are related but NOT the same.

3.4 Foundry Tools (vs Tools in Responses API)

Important: Foundry Tools are Azure AI services, NOT the same as "tools" in the Responses API.

Service / Tool	Role	Best For
Azure Language	Text analysis, entity extraction, sentiment, summarisation, question answering	Analyzing text documents and feedback
Azure Speech	Speech-to-text, text-to-speech, live voice for conversational apps	Voice input/output for applications
Azure Translator	State-of-the-art language translation	Multi-language support
Azure Document Intelligence	Extract fields from documents/forms (invoices, receipts, contracts)	Structured document extraction
Azure Content Understanding	Multimodal content analysis (documents, images, audio, video)	Broader content extraction scenarios

How to use: Create client applications connecting to tool-specific endpoint in Foundry resource using APIs and SDKs.

4. Azure Services / Tools

Service / Tool	Role	Exam Relevance
Microsoft Foundry	Central platform for Azure AI development	High

Service / Tool	Role	Exam Relevance
Foundry portal	Create, test, deploy, and manage projects, models, agents, tools	High
Foundry SDK	Programmatic project and asset management	Medium-High
Foundry Models	Model catalog and deployments	High
Foundry Agent Service	Build and consume agents	High
Foundry Tools	Prebuilt AI APIs/models for common tasks	High
Foundry IQ	Central MCP-based knowledge connection for agents	Medium-High
MCP connections	Standardised tool/knowledge integration	Medium-High
Azure Language	Text analysis, entity extraction, sentiment, summarisation	Medium
Azure Speech	Speech-to-text, text-to-speech, live voice	Medium
Azure Translator	Text translation	Medium
Azure Document Intelligence	Extract fields from documents/forms	Medium-High
Azure Content Understanding	Multimodal content extraction from documents, images, audio, video	Medium-High
VS Code Foundry Toolkit	Browse/manage projects, models, agents, connections, vector stores	Medium
Project endpoint	Access Foundry models, Foundry APIs, Agent Service	High

Service / Tool	Role	Exam Relevance
Azure OpenAI endpoint	Use OpenAI-compatible APIs such as Chat Completions and Responses API	High
Microsoft Entra ID	Preferred production authentication	High

5. Decision Rules

Scenario	Choose	Why
Full Azure AI app or agent solution	Microsoft Foundry project	Best project/resource organisation
Very simple isolated AI service use case	Individual AI resource may be enough	Lower overhead
Need model-based chat app	Foundry model deployment	Gives endpoint for app integration
Need autonomous or semi-autonomous task execution	Foundry Agent Service	Agents combine model, instructions, and tools
Need external/custom tool access	MCP or custom tools	Standardised tool connectivity
Need shared knowledge grounding for agents	Foundry IQ	Central knowledge connection
Need entity/sentiment/summarisation	Azure Language	Purpose-built text analysis
Need speech input/output	Azure Speech	Purpose-built speech service
Need document field extraction	Document Intelligence	Better for forms, invoices, receipts
Need multimodal extraction	Content Understanding	Works across documents, images, audio, video
Need production authentication	Microsoft Entra ID	Better than key-only authentication
Need quick no-code testing	Foundry portal playground	Validate model/agent behaviour quickly
Need IDE-based development	VS Code + Foundry Toolkit	Developer productivity

6. Responsible AI Principles

Microsoft's six core principles for responsible AI:

Principle	Meaning	Example
Fairness	Treat all people fairly; avoid bias based on gender, ethnicity, etc.	Loan approval model treats all applicants equally
Reliability and Safety	Perform reliably and safely; minimize risk to human life	Autonomous vehicle AI has rigorous testing
Privacy and Security	Respect privacy; protect personal data and customer content	Encrypt data, implement access controls
Inclusiveness	Empower everyone; bring benefits to all parts of society	Design tested by diverse user groups
Transparency	Be understandable; communicate capabilities and limitations	Tell users how the system works and its limits
Accountability	People are responsible for AI systems; ensure governance and standards	Define accountability frameworks and monitoring

Exam tip: System messages influence behavior but **don't guarantee** compliance. Always combine with testing, evaluation, and guardrails.

7. Configuration / Implementation Notes

Foundry Project Setup

Basic flow:

1. Open Microsoft Foundry portal (<https://ai.azure.com>)
2. Create a new project with unique name
3. Select subscription, resource group, region
4. Foundry resource: Use default name (usually {project_name}-resource)
5. Deploy a model from the model catalog
6. Test the model in the playground
7. Review resource-level and project-level details

- 8. Note project endpoint, Azure OpenAI endpoint, and keys
- 9. Install and connect VS Code Foundry Toolkit
- 10. Clean up resources to avoid cost

Authentication

Method	Use	Best Practice
Key-based authentication	Simple development/testing	Development only
Microsoft Entra ID / token-based authentication	Preferred for production	Always use in production

8. Exam Traps

Trap	Correct Understanding
Thinking Foundry is only a model playground	Foundry is a full AI development platform
Confusing Foundry resource with Foundry project	Resource = Azure shared infrastructure; Project = solution workspace
Assuming every task needs a generative AI agent	Foundry Tools can be cheaper, more predictable, and purpose-built
Using keys as the default production security model	Entra ID/token-based authentication is preferred for production
Confusing project endpoint and Azure OpenAI endpoint	They support different integration patterns
Ignoring responsible AI until deployment	Responsible AI must be considered from planning through operation
Thinking agents are just chatbots	Agents combine model, instructions, tools, knowledge, and task responsibility
Treating Document Intelligence and Content Understanding as identical	Document Intelligence is document/form extraction; Content Understanding is broader multimodal
Assuming classic Foundry hub architecture is the latest model	This module focuses on the newer Foundry project architecture
Confusing Foundry Tools with tools in Responses API	Foundry Tools = Azure services; Responses API tools = code_interpreter, web_search, file_search, function

9. Hands-on Checklist

By the end of this module, you should be able to:

- * [] Explain common Azure AI capability areas with examples
- * [] Explain what Microsoft Foundry is used for
- * [] Distinguish between a Foundry resource and a Foundry project
- * [] Identify the role of models, agents, tools, and knowledge in Foundry
- * [] Choose when to use Foundry Tools instead of an LLM-only approach
- * [] Identify the main Foundry Tools and their use cases
- * [] Explain the purpose of the project endpoint and Azure OpenAI endpoint
- * [] Create a Foundry project
- * [] Deploy and test a model in the Foundry playground
- * [] Install and use the VS Code Foundry Toolkit
- * [] Explain why Entra ID is preferred for production authentication
- * [] Recall Microsoft's six responsible AI principles with examples
- * [] Clean up Azure resources to avoid unnecessary cost

10. Quick Revision Questions

1. What is Microsoft Foundry used for?
2. What is the difference between a Foundry resource and a Foundry project?
3. When would you use Foundry Tools instead of relying only on a generative AI model?
4. What is the role of an agent in Microsoft Foundry?
5. What does Foundry IQ provide?
6. What is the difference between the project endpoint and the Azure OpenAI endpoint?
7. Which authentication method is preferred for production: keys or Entra ID?
8. Which Foundry Tool would you use for extracting fields from invoices and receipts?
9. Which Foundry Tool would you use for multimodal analysis across documents, images, audio, and video?
10. Which Foundry Tool supports speech-to-text and text-to-speech?
11. What are the six Microsoft responsible AI principles?
12. Why should responsible AI be considered during planning rather than only after deployment?
13. What can the VS Code Foundry Toolkit help you do?
14. Why is a model card useful before deploying a model?
15. Why should lab resources be deleted after completing the exercise?

Final Exam Memory Hook

Plan first:

Capability → Foundry Project → Model/Agent/Tool → Endpoint → Auth → Safety → Monitor → Cost

One-line summary: Use Microsoft Foundry to organise, build, test, secure, and operate Azure AI apps and agents; choose specialised Foundry Tools where they are more predictable than an LLM-only design.

Key Enhancements in This Version

- OK Added capability examples column
- OK Added Foundry Tools with use cases table
- OK Added critical distinction between Foundry Tools and Responses API tools
- OK Added responsible AI principles with examples
- OK Enhanced decision rules with more scenarios
- OK Clarified exam traps with specific distinctions